Worksheet 29 · Transformations of Shapes IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Describe and perform reflections, rotations and translations • Describe and perform enlargements, including fractional scale factors • Describe a single transformation fully		
Grade: / 57	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

Translation → By vector $\begin{pmatrix} a \\ b \end{pmatrix}$ a right −a left; b up −b down

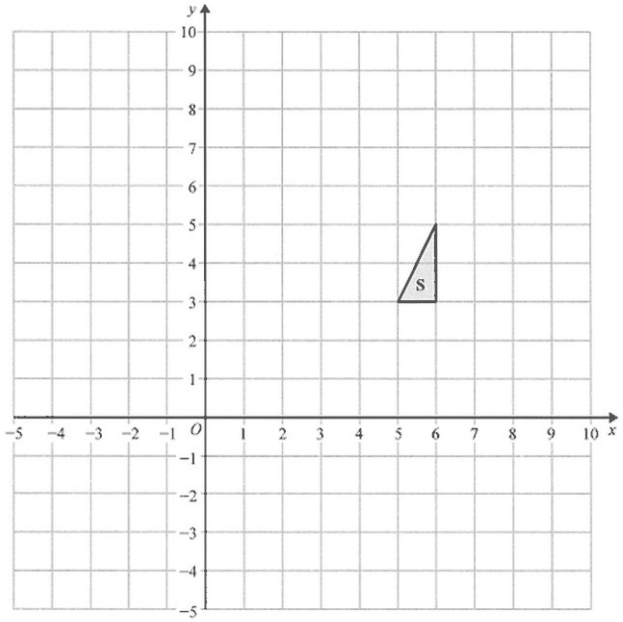
Reflection → In line $x=a$ (horizontal) $y=b$ (vertical)

Rotation → Centre, angle, direction (CW or ACW)

Enlargement → Centre, Scale Factor

Questions *Transformations of Shapes*

Question 1 (3 marks)



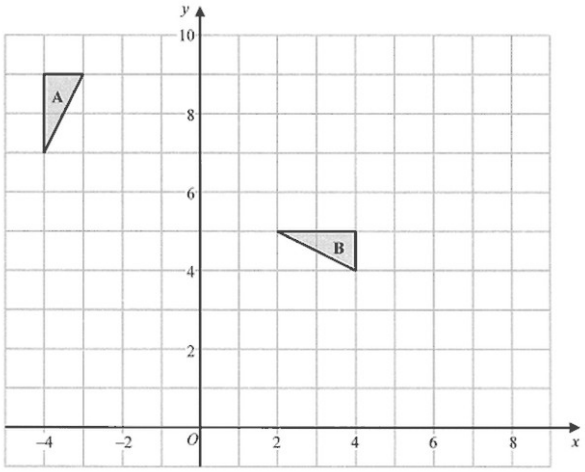
(a) Reflect triangle S in the line $y = x$
Label the new triangle R.

(2)

(b) Translate triangle S by the vector $\begin{pmatrix} -4 \\ -6 \end{pmatrix}$
Label the new triangle T.

(1)

Question 2 (4 marks)



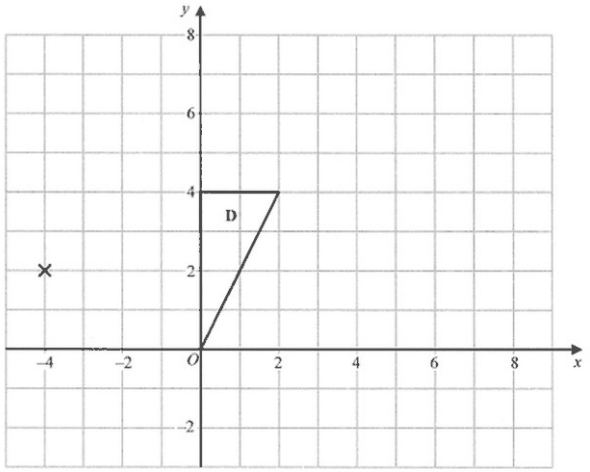
(a) Describe fully the single transformation that maps triangle A onto triangle B.

(3)

(b) On the grid, translate triangle A by the vector $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$
Label the new triangle C.

(1)

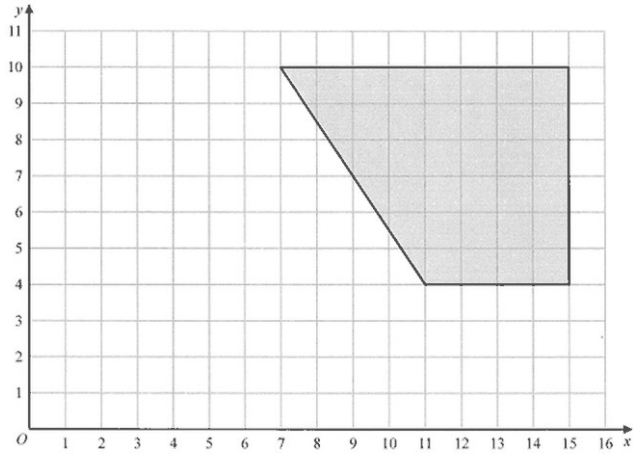
Question 3 (6 marks)



(c) On the grid, enlarge triangle **D** with scale factor $\frac{1}{2}$ and centre (−4, 2)

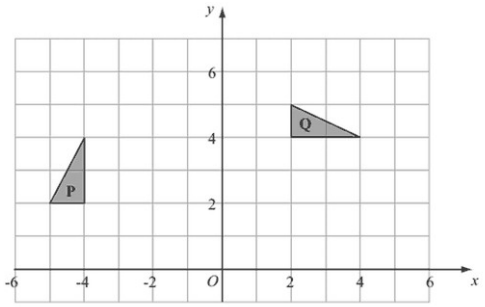
(2)

Question 4 (2 marks)



On the grid, enlarge the shaded shape with scale factor $\frac{1}{2}$ and centre (1, 2)

Question 5 (5 marks)



(a) Describe fully the single transformation which maps triangle **P** onto triangle **Q**.

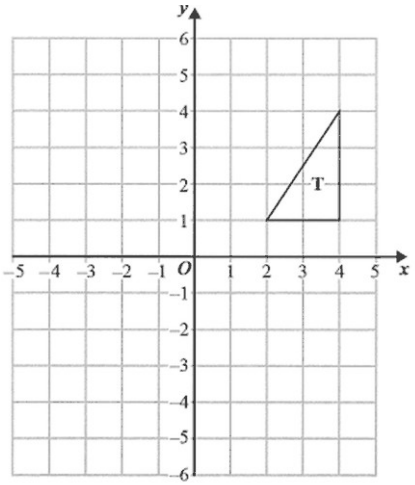
(3)

(b) Reflect triangle **Q** in the line $y = x$.

Label the new triangle **R**.

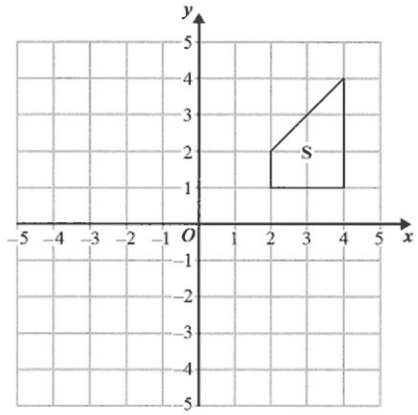
(2)

Question 6 (3 marks)



(*a*) On the grid above, rotate triangle **T** 90° clockwise about (0, 2).

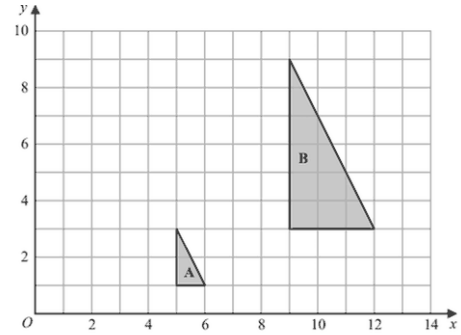
(2)



(*b*) On the grid, translate shape **S** by the vector $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$.

(1)

Question 7 (3 marks)

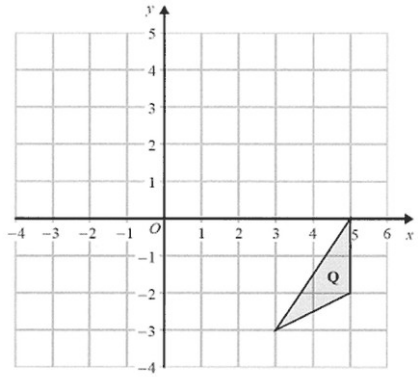


(a) Describe fully the single transformation that maps triangle **A** onto triangle **B**

(3)

(b) On the grid above, translate triangle **A** by the vector $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$

Question 8 (5 marks)



(*a*) On the grid, reflect triangle **Q** in the line $x = 1$
Label the new triangle **R**.

(2)

Triangle **R** is mapped onto triangle **S** by a reflection in the line $y = 0$

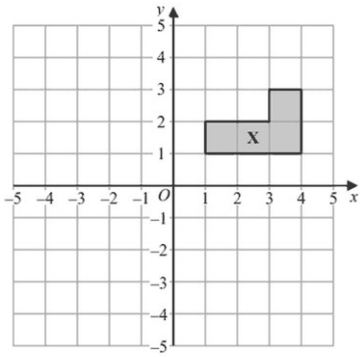
(*b*) Describe fully the single transformation that maps triangle **Q** onto triangle **S**.

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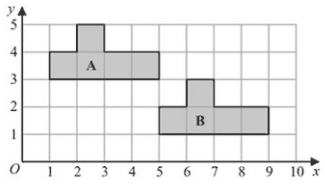
(3)

Question 9 (4 marks)



(a) On the grid above, rotate shape **X** 90° clockwise about *O*

(2)



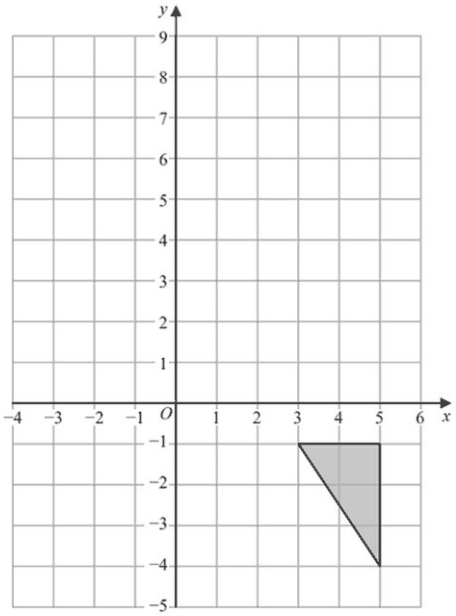
(b) Describe fully the single transformation that maps shape **A** onto shape **B**

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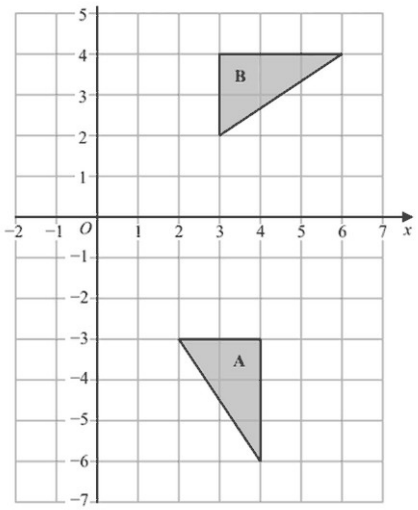
(2)

Question 10 (5 marks)



(a) On the grid, reflect the shaded triangle in the line with equation $y = 2$

(2)

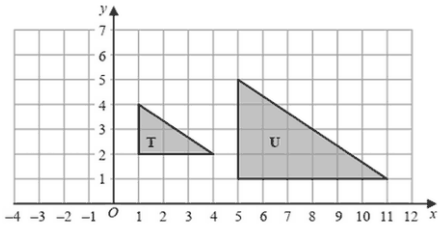


(b) Describe fully the single transformation that maps triangle **A** onto triangle **B**.

(3)

Question 11 (3 marks)

Here are triangle **T** and triangle **U** drawn on a grid of squares.



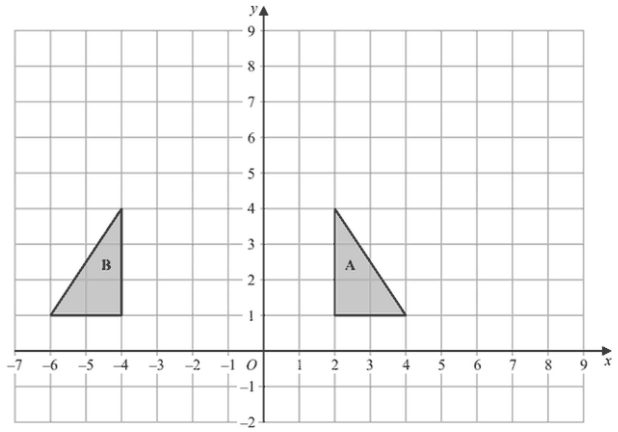
(c) Describe fully the single transformation that maps triangle **T** onto triangle **U**.

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(3)

Question 12 (2 marks)



(a) Describe fully the single transformation that maps triangle **A** onto triangle **B**

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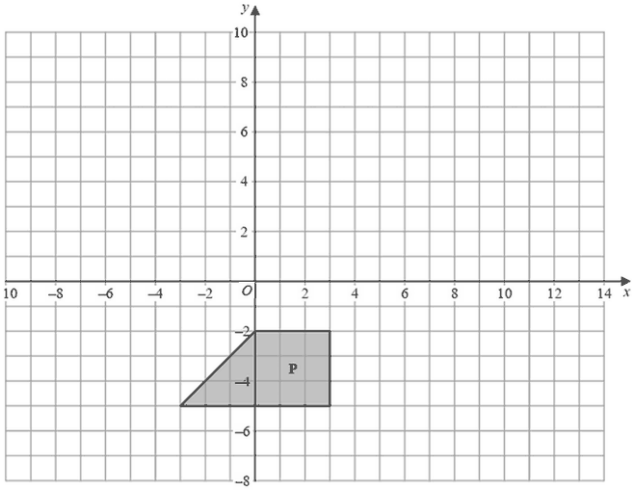
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(2)

(b) On the grid above, enlarge triangle **A** with scale factor 2 and centre O

Question 13 (3 marks)

Here is a shape **P** drawn on a grid of squares.



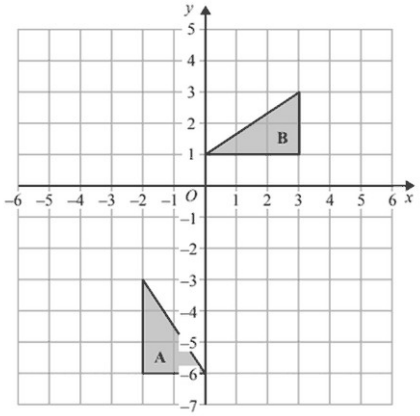
(a) On the grid, rotate shape **P** 180° about the point (–3, 2)
Label the new shape **Q**.

(2)

(b) On the grid, translate shape **P** by the vector $\begin{pmatrix} 10 \\ 8 \end{pmatrix}$
Label the new shape **R**.

(1)

Question 14 (4 marks)



(a) On the grid, translate triangle **A** by the vector $\begin{pmatrix} 5 \\ 2 \end{pmatrix}$

(1)

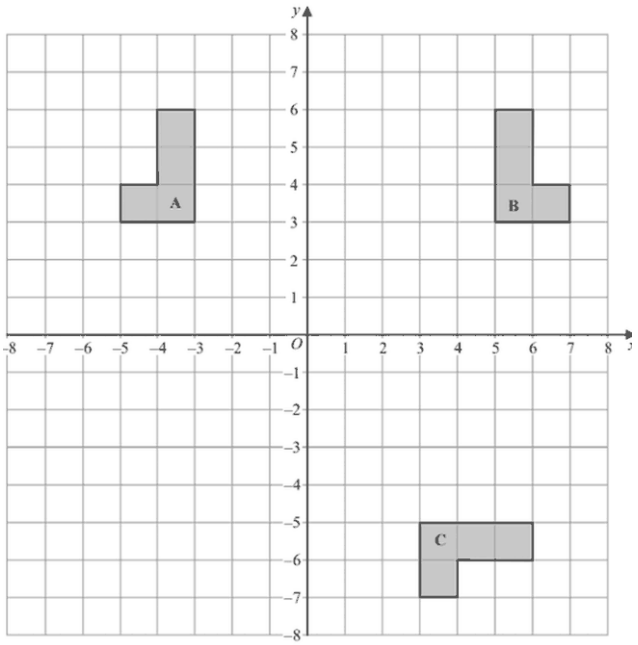
(b) Describe fully the single transformation that maps triangle **A** onto triangle **B**.

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(3)

Question 15 (5 marks)



(a) Describe fully the single transformation that maps shape **A** onto shape **B**.

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(2)

(b) Describe fully the single transformation that maps shape **B** onto shape **C**.

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(3)